

Exploratory Analysis

2023-12-05

R Markdown

This is an R Markdown document. Markdown is a simple formatting syntax for authoring HTML, PDF, and MS Word documents. For more details on using R Markdown see <http://rmarkdown.rstudio.com>.

When you click the **Knit** button a document will be generated that includes both content as well as the output of any embedded R code chunks within the document. You can embed an R code chunk like this:

```
#set up libraries  
library(caret)
```

```
## Loading required package: ggplot2
```

```
## Loading required package: lattice
```

```
library(ggplot2)  
library('naivebayes')
```

```
## Warning: package 'naivebayes' was built under R version 4.3.2
```

```
## naivebayes 0.9.7 loaded
```

```
library('C50')
```

```
## Warning: package 'C50' was built under R version 4.3.2
```

```
#Open file  
getwd()
```

```
## [1] "C:/Users/Grant/OneDrive/Documents/SCHOOL/DTSC3010"
```

```
setwd("C:/Users/Grant/OneDrive/Documents/RFiles")  
nbadf <- read.csv('NBAPlayerData.csv')
```

```
#statistical summary of data
```

```
summary(nbadf)
```

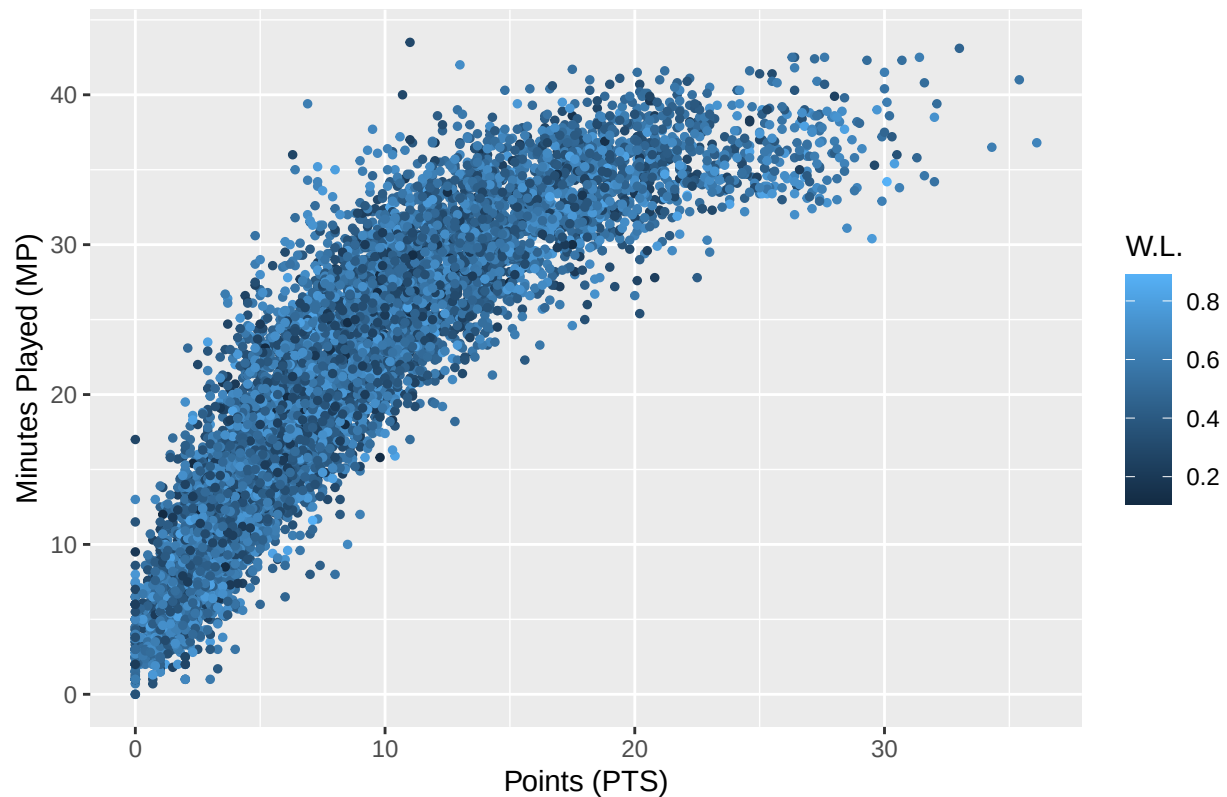
##	XO	Player	Pos	Age
##	Min. : 0	Length:9660	Length:9660	Min. :18.00
##	1st Qu.:2415	Class :character	Class :character	1st Qu.:23.00
##	Median :4830	Mode :character	Mode :character	Median :26.00
##	Mean :4830			Mean :26.39
##	3rd Qu.:7244			3rd Qu.:29.00
##	Max. :9659			Max. :44.00
##	Tm	G	GS	MP
##	Length:9660	Min. : 1.00	Min. : 0.0	Min. : 0.00
##	Class :character	1st Qu.:31.00	1st Qu.: 1.0	1st Qu.:12.20
##	Mode :character	Median :58.00	Median :10.0	Median :19.60
##		Mean :51.36	Mean :24.8	Mean :20.18
##		3rd Qu.:73.00	3rd Qu.:49.0	3rd Qu.:28.20
##		Max. :85.00	Max. :83.0	Max. :43.50
##	FG	FGA	FG.	X3P
##	Min. : 0.000	Min. : 0.000	Min. :0.0000	Min. :0.0000
##	1st Qu.: 1.400	1st Qu.: 3.300	1st Qu.:0.3990	1st Qu.:0.0000
##	Median : 2.600	Median : 5.800	Median :0.4380	Median :0.4000
##	Mean : 3.078	Mean : 6.855	Mean :0.4369	Mean :0.6553
##	3rd Qu.: 4.300	3rd Qu.: 9.500	3rd Qu.:0.4830	3rd Qu.:1.1000
##	Max. :12.200	Max. :27.200	Max. :1.0000	Max. :5.3000
##	X3PA	X3P.	X2P	X2PA
##	Min. : 0.000	Min. :0.0000	Min. : 0.000	Min. : 0.000
##	1st Qu.: 0.100	1st Qu.:0.0000	1st Qu.: 1.000	1st Qu.: 2.200
##	Median : 1.300	Median :0.3130	Median : 1.900	Median : 4.000
##	Mean : 1.874	Mean :0.2518	Mean : 2.424	Mean : 4.982
##	3rd Qu.: 3.100	3rd Qu.:0.3680	3rd Qu.: 3.400	3rd Qu.: 6.900
##	Max. :13.200	Max. :1.0000	Max. :10.400	Max. :22.200
##	X2P.	eFG.	FT	FTA
##	Min. :0.0000	Min. :0.000	Min. : 0.000	Min. : 0.000
##	1st Qu.:0.4310	1st Qu.:0.448	1st Qu.: 0.500	1st Qu.: 0.700
##	Median :0.4760	Median :0.491	Median : 1.000	Median : 1.400
##	Mean :0.4703	Mean :0.481	Mean : 1.455	Mean : 1.931
##	3rd Qu.:0.5220	3rd Qu.:0.530	3rd Qu.: 1.900	3rd Qu.: 2.600
##	Max. :1.0000	Max. :1.500	Max. :10.200	Max. :11.800
##	FT.	ORB	DRB	TRB
##	Min. :0.0000	Min. :0.0000	Min. : 0.000	Min. : 0.000
##	1st Qu.:0.6470	1st Qu.:0.3000	1st Qu.: 1.400	1st Qu.: 1.800
##	Median :0.7500	Median :0.7000	Median : 2.300	Median : 3.000
##	Mean :0.7016	Mean :0.9172	Mean : 2.644	Mean : 3.559
##	3rd Qu.:0.8190	3rd Qu.:1.3000	3rd Qu.: 3.500	3rd Qu.: 4.700
##	Max. :1.0000	Max. :5.5000	Max. :11.400	Max. :16.000
##	AST	STL	BLK	TOV
##	Min. : 0.000	Min. :0.0000	Min. :0.0000	Min. :0.00
##	1st Qu.: 0.600	1st Qu.:0.3000	1st Qu.:0.1000	1st Qu.:0.60
##	Median : 1.200	Median :0.6000	Median :0.3000	Median :1.00
##	Mean : 1.818	Mean :0.6314	Mean :0.4055	Mean :1.15
##	3rd Qu.: 2.400	3rd Qu.:0.9000	3rd Qu.:0.5000	3rd Qu.:1.50
##	Max. :11.700	Max. :2.9000	Max. :3.7000	Max. :5.70
##	PF	PTS	Year	Pts.Won
##	Min. :0.000	Min. : 0.000	Min. :2003	Min. : 0.000
##	1st Qu.:1.200	1st Qu.: 3.700	1st Qu.:2008	1st Qu.: 0.000
##	Median :1.800	Median : 6.800	Median :2013	Median : 0.000
##	Mean :1.795	Mean : 8.264	Mean :2013	Mean : 6.315

```
## 3rd Qu.:2.400 3rd Qu.:11.600 3rd Qu.:2018 3rd Qu.: 0.000
## Max. :6.000 Max. :36.100 Max. :2022 Max. :1310.000
## Pts.Max Share Team W
## Min. : 0.00 Min. :0.000000 Length:9660 Min. : 7.00
## 1st Qu.: 0.00 1st Qu.:0.000000 Class :character 1st Qu.:31.00
## Median : 0.00 Median :0.000000 Mode :character Median :41.00
## Mean : 33.58 Mean :0.005382 Mean :39.84
## 3rd Qu.: 0.00 3rd Qu.:0.000000 3rd Qu.:49.00
## Max. :1310.00 Max. :1.000000 Max. :73.00
## L W.L. GB PS.G
## Min. : 9.00 Min. :0.1060 Min. : 0.00 Min. : 84.2
## 1st Qu.:31.00 1st Qu.:0.3900 1st Qu.: 3.00 1st Qu.: 96.9
## Median :40.00 Median :0.5120 Median :12.00 Median :101.4
## Mean :40.19 Mean :0.4976 Mean :14.59 Mean :102.2
## 3rd Qu.:49.00 3rd Qu.:0.6100 3rd Qu.:24.00 3rd Qu.:107.0
## Max. :72.00 Max. :0.8900 Max. :56.00 Max. :120.1
## PA.G SRS
## Min. : 84.3 Min. : -13.96000
## 1st Qu.: 97.0 1st Qu.: -3.12000
## Median :101.5 Median : -0.04000
## Mean :102.3 Mean : -0.08058
## 3rd Qu.:107.3 3rd Qu.: 3.12250
## Max. :119.7 Max. : 11.35000
```

##visualization plot #1: scatterplot for points per game on x axis and Minutes played ##on the y axis.
Colored by team

```
ggplot(data = nbadf)+
  geom_point(mapping = aes(x = PTS , y = MP, color = W.L.), size = 1) +
  labs(title = "NBA Player Points vs Minutes Played",
        x = "Points (PTS)",
        y = "Minutes Played (MP)")
```

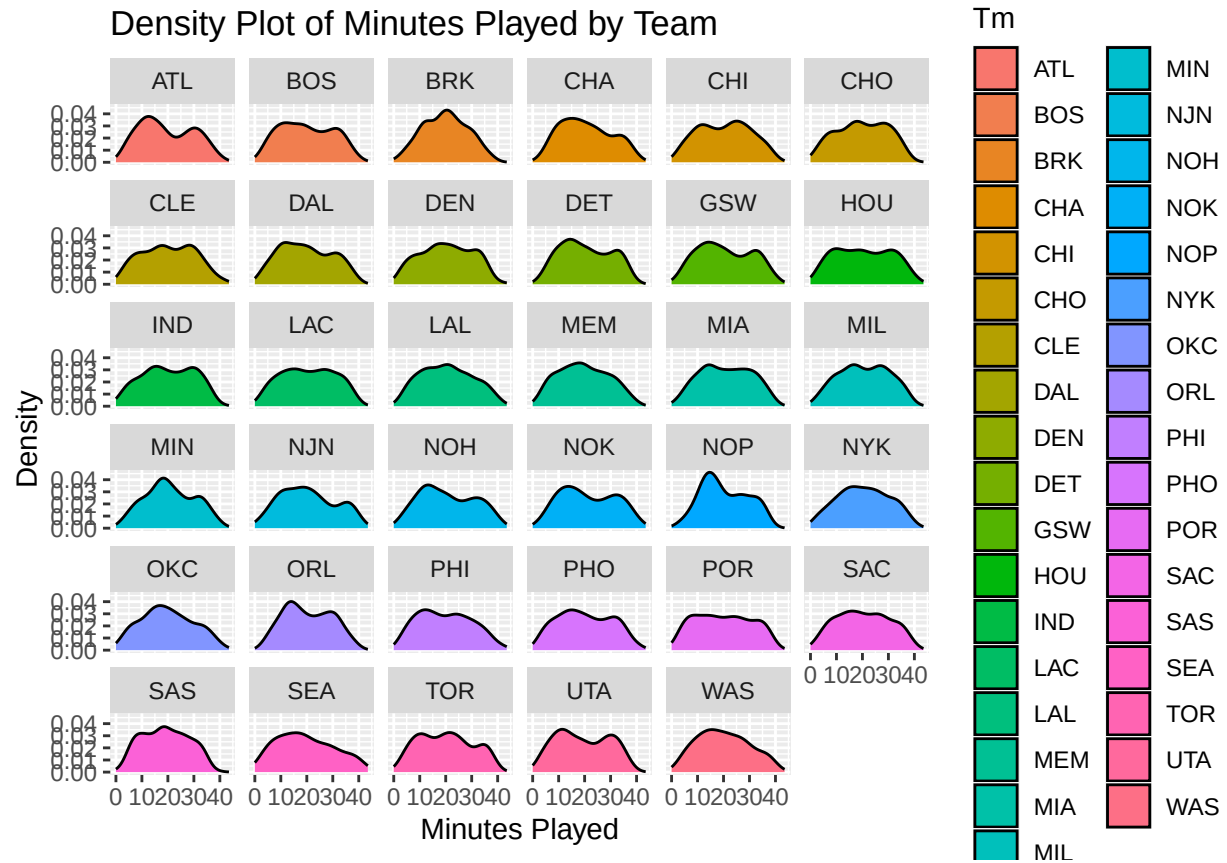
NBA Player Points vs Minutes Played



##visualization plot #2: Density plot of minutes played per team

```
ggplot(nbadf, aes(x = MP, fill = Tm)) +  
  geom_density(aes(y = ..density..)) +  
  facet_wrap(~ Tm) +  
  labs(title = "Density Plot of Minutes Played by Team", x = "Minutes Played", y = "Density")
```

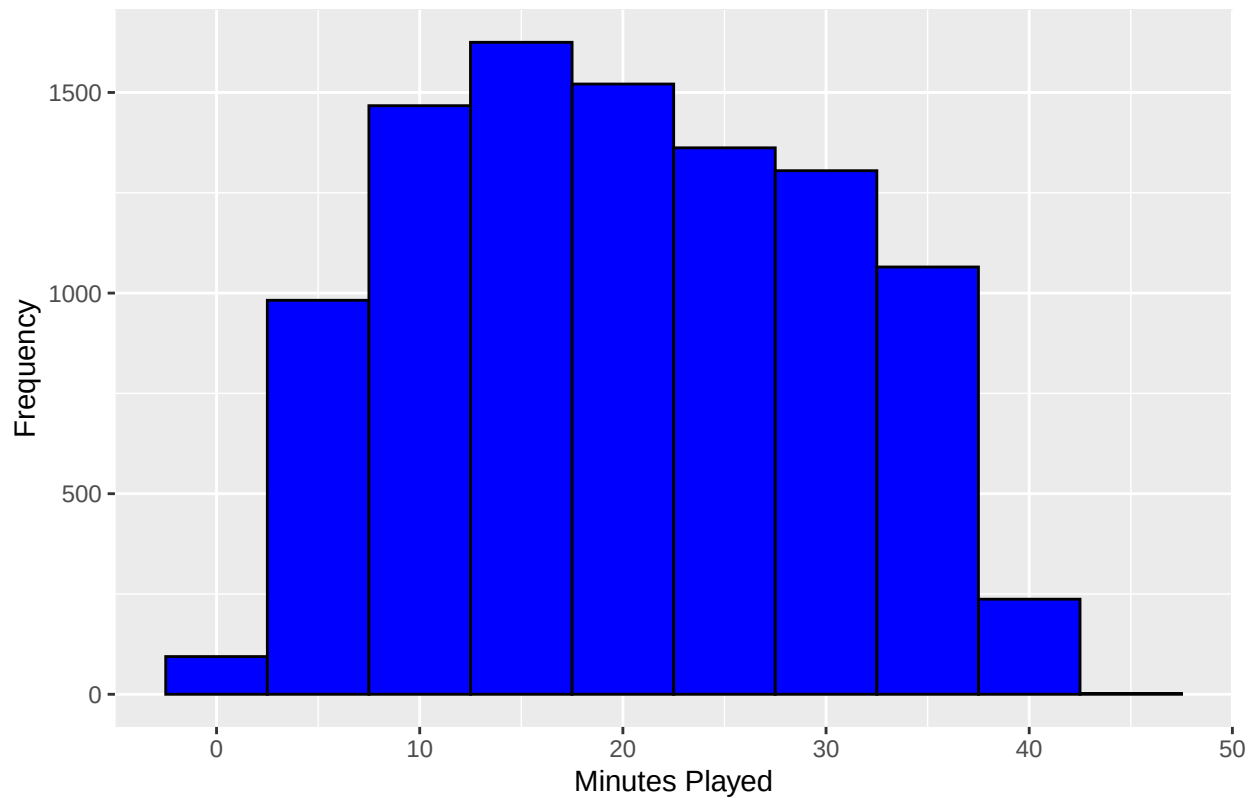
```
## Warning: The dot-dot notation ('..density..') was deprecated in ggplot2 3.4.0.  
## i Please use 'after_stat(density)' instead.  
## This warning is displayed once every 8 hours.  
## Call 'lifecycle::last_lifecycle_warnings()' to see where this warning was  
## generated.
```



##visualization plot #3: Histogram showing the distribution of minutes played throughout the league

```
ggplot(nbadf, aes(x = MP)) +
  geom_histogram(binwidth = 5, fill = "blue", color = "black") +
  labs(title = "Distribution of Minutes Played", x = "Minutes Played", y = "Frequency")
```

Distribution of Minutes Played



#statistical test #1:

```
teamA <- nbadf[nbadf$Tm %in% c("LAL"), ]$MP
teamB <- nbadf[nbadf$Tm %in% c("WAS"), ]$MP
t.test(teamA, teamB)
```

```
##
## Welch Two Sample t-test
##
## data: teamA and teamB
## t = 0.68958, df = 658.66, p-value = 0.4907
## alternative hypothesis: true difference in means is not equal to 0
## 95 percent confidence interval:
## -0.9709533 2.0220658
## sample estimates:
## mean of x mean of y
## 20.59083 20.06527
```

#statistical test #2:

```
nbadf$GB <- as.factor(nbadf$GB)
nbadf$W.L. <- as.factor(nbadf$W.L.)

# Perform two-way ANOVA
result_anova <- aov(MP ~ GB + W.L., data = nbadf)
print(result_anova)
```

```
## Call:
##   aov(formula = MP ~ GB + W.L., data = nbadf)
##
## Terms:
##              GB      W.L. Residuals
## Sum of Squares 3395.1  3526.8 922002.0
## Deg. of Freedom   60    111    9488
##
## Residual standard error: 9.857768
## 4 out of 176 effects not estimable
## Estimated effects may be unbalanced
```